

Example 7 expresses $(\frac{\partial U}{\partial T})_P$ in terms of three other partial derivatives. Two of them, namely $(\frac{\partial U}{\partial T})_V$, the constant-volume heat capacity, and $(\frac{\partial V}{\partial T})_P$, the expansion coefficient, can be easily measured experimentally. The third, the internal pressure, $(\frac{\partial U}{\partial V})_T$, cannot be measured directly but can be related to $(\frac{\partial P}{\partial T})_V$, which is measurable. Thus, $(\frac{\partial U}{\partial T})_P$ can be determined indirectly using this identity.

Exercises and Problems for Section 14.6

Exercises

For Exercises 1–6, find dz/dt using the chain rule. Assume the variables are restricted to domains on which the functions are defined.

1. $z = xy^2$, $x = e^{-t}$, $y = \sin t$
2. $z = x \sin y + y \sin x$, $x = t^2$, $y = \ln t$
3. $z = \sin(x/y)$, $x = 2t$, $y = 1 - t^2$
4. $z = \ln(x^2 + y^2)$, $x = 1/t$, $y = \sqrt{t}$
5. $z = xe^y$, $x = 2t$, $y = 1 - t^2$
6. $z = (x + y)e^y$, $x = 2t$, $y = 1 - t^2$

For Exercises 7–15, find $\partial z/\partial u$ and $\partial z/\partial v$. The variables are restricted to domains on which the functions are defined.

7. $z = \sin(x/y)$, $x = \ln u$, $y = v$
8. $z = \ln(xy)$, $x = (u^2 + v^2)^2$, $y = (u^3 + v^3)^2$
9. $z = xe^y$, $x = \ln u$, $y = v$
10. $z = (x + y)e^y$, $x = \ln u$, $y = v$
11. $z = xe^y$, $x = u^2 + v^2$, $y = u^2 - v^2$
12. $z = (x + y)e^y$, $x = u^2 + v^2$, $y = u^2 - v^2$
13. $z = xe^{-y} + ye^{-x}$, $x = u \sin v$, $y = v \cos u$
14. $z = \cos(x^2 + y^2)$, $x = u \cos v$, $y = u \sin v$
15. $z = \tan^{-1}(x/y)$, $x = u^2 + v^2$, $y = u^2 - v^2$

Problems

16. Use the chain rule to find dz/dt , and check the result by expressing z as a function of t and differentiating directly.

$$z = x^3 y^2, \quad x = t^3, \quad y = t^2$$

17. Use the chain rule to find $\partial w/\partial \rho$ and $\partial w/\partial \theta$, given that

$$w = x^2 + y^2 - z^2,$$

and

$$x = \rho \sin \phi \cos \theta, \quad y = \rho \sin \phi \sin \theta, \quad z = \rho \cos \phi.$$

18. A bison is charging across the plain one morning. His path takes him to location (x, y) at time t where x and y are functions of t and north is in the direction of increasing y . The temperature is always colder farther north. As time passes, the sun rises in the sky, sending out more heat, and a cold front blows in from the east. At time t the air temperature H near the bison is given by $H = f(x, y, t)$. The chain rule expresses the derivative dH/dt as a sum of three terms:

$$\frac{dH}{dt} = \frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt} + \frac{\partial f}{\partial t}.$$

Identify the term that gives the contribution to the change in temperature experienced by the bison that is due to

- (a) The rising sun.
- (b) The coming cold front.
- (c) The bison's change in latitude.

19. Let $z = f(x, y)$ where $x = g(t)$, $y = h(t)$ and f, g, h are all differentiable functions. Given the information in the table, find $\frac{\partial z}{\partial t} \Big|_{t=1}$.

$f(3, 10) = 7$	$f(4, 11) = -20$
$f_x(3, 10) = 100$	$f_y(3, 10) = 0.1$
$f_x(4, 11) = 200$	$f_y(4, 11) = 0.2$
$f(3, 4) = -10$	$f(10, 11) = -1$
$g(1) = 3$	$h(1) = 10$
$g'(1) = 4$	$h'(1) = 11$

20. The voltage, V , (in volts) across a circuit is given by Ohm's law: $V = IR$, where I is the current (in amps) flowing through the circuit and R is the resistance (in ohms). If we place two circuits, with resistance R_1 and R_2 , in parallel, then their combined resistance, R , is given by

$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2}.$$

Suppose the current is 2 amps and increasing at 10^{-2} amp/sec and R_1 is 3 ohms and increasing at 0.5 ohm/sec, while R_2 is 5 ohms and decreasing at 0.1 ohm/sec. Calculate the rate at which the voltage is changing.

21. The air pressure is decreasing at a rate of 2 pascals per kilometer in the eastward direction. In addition, the air pressure is dropping at a constant rate with respect to time everywhere. A ship sailing eastward at 10 km/hour past an island takes barometer readings and records a pressure drop of 50 pascals in 2 hours. Estimate the time rate of change of air pressure on the island. (A pascal is a unit of air pressure.)
22. A steel bar with square cross sections 5 cm by 5 cm and length 3 meters is being heated. For each dimension, the bar expands $13 \cdot 10^{-6}$ meters for each 1°C rise in temperature.² What is the rate of change in the volume of the steel bar?
23. Corn production, C , is a function of rainfall, R , and temperature, T . (See Example 1 on page 796.) Figures 14.52 and 14.53 show how rainfall and temperature are predicted to vary with time because of global warming. Suppose we know that $\Delta C \approx 3.3\Delta R - 5\Delta T$. Use this to estimate the change in corn production between the year 2020 and the year 2021. Hence, estimate dC/dt when $t = 2020$.

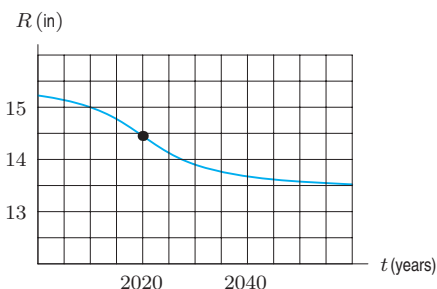


Figure 14.52: Rainfall as a function of time

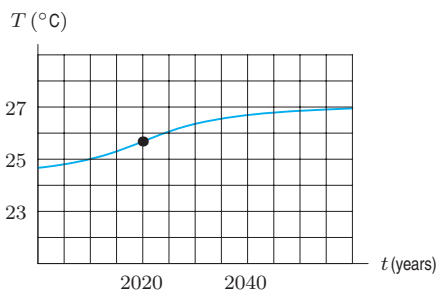


Figure 14.53: Temperature as a function of time

24. The function $g(\rho)$ is graphed in Figure 14.54. Let $\rho = \sqrt{x^2 + y^2 + z^2}$. Define f , a function of x, y, z by $f(x, y, z) = g(\sqrt{x^2 + y^2 + z^2})$. Let $\vec{F} = \text{grad } f$.
- (a) Describe precisely in words the level surfaces of f .

- (b) Give a unit vector in the direction of $\vec{F}$ at the point $(1, 2, 2)$.
- (c) Estimate $\|\vec{F}\|$ at the point $(1, 2, 2)$.
- (d) Estimate $\vec{F}$ at the point $(1, 2, 2)$.
- (e) The points $(1, 2, 2)$ and $(3, 0, 0)$ are both on the sphere $x^2 + y^2 + z^2 = 9$. Estimate $\vec{F}$ at $(3, 0, 0)$.
- (f) If P and Q are any two points on the sphere $x^2 + y^2 + z^2 = k^2$:
- Compare the magnitudes of $\vec{F}$ at P and at Q .
 - Describe the directions of $\vec{F}$ at P and at Q .

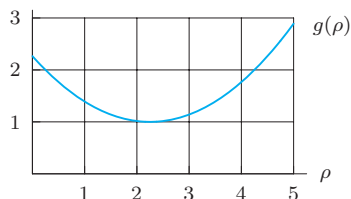


Figure 14.54

25. Let $z = g(u, v, w)$ and $u = u(s, t)$, $v = v(s, t)$, $w = w(s, t)$. How many terms are there in the expression for $\partial z / \partial t$?
26. Suppose $w = f(x, y, z)$ and that x, y, z are functions of u and v . Use a tree diagram to write down the chain rule formula for $\partial w / \partial u$ and $\partial w / \partial v$.
27. Suppose $w = f(x, y, z)$ and that x, y, z are all functions of t . Use a tree diagram to write down the chain rule for dw/dt .
28. Let $F(u, v)$ be a function of two variables. Find $f'(x)$ if
- $f(x) = F(x, 3)$
 - $f(x) = F(3, x)$
 - $f(x) = F(x, x)$
 - $f(x) = F(5x, x^2)$

In Problems 29–30, let $z = f(x, y)$, $x = x(u, v)$, $y = y(u, v)$ and $x(1, 2) = 5$, $y(1, 2) = 3$, calculate the partial derivative in terms of some of the numbers a, b, c, d, e, k, p, q :

$$\begin{aligned} f_x(1, 2) &= a & f_y(1, 2) &= c & x_u(1, 2) &= e & y_u(1, 2) &= p \\ f_x(5, 3) &= b & f_y(5, 3) &= d & x_v(1, 2) &= k & y_v(1, 2) &= q \end{aligned}$$

29. $z_u(1, 2)$

30. $z_v(1, 2)$

31. The equation $f(x, y) = f(a, b)$ defines a level curve through a point (a, b) where $\text{grad } f(a, b) \neq \vec{0}$. Use implicit differentiation and the chain rule to show that the slope of the line tangent to this curve at the point (a, b) is $-f_x(a, b)/f_y(a, b)$ if $f_y(a, b) \neq 0$.
32. Let $z = f(t)g(t)$. Use the chain rule applied to $h(x, y) = f(x)g(y)$ to show that $dz/dt = f'(t)g(t) + f(t)g'(t)$. The one-variable product rule for differentiation is a special case of the two-variable chain rule.

²<http://www.engineeringtoolbox.com/>

33. A function $f(x, y)$ is *homogeneous of degree p* if $f(tx, ty) = t^p f(x, y)$ for all t . Show that any differentiable, homogeneous function of degree p satisfies Euler's Theorem:

$$x f_x(x, y) + y f_y(x, y) = p f(x, y).$$

[Hint: Define $g(t) = f(tx, ty)$ and compute $g'(1)$.]

34. Let $F(x, y, z)$ be a function and define a function $z = f(x, y)$ implicitly by letting $F(x, y, f(x, y)) = 0$. Use the chain rule to show that

$$\frac{\partial z}{\partial x} = -\frac{\partial F/\partial x}{\partial F/\partial z} \quad \text{and} \quad \frac{\partial z}{\partial y} = -\frac{\partial F/\partial y}{\partial F/\partial z}.$$

In Problems 35–36, let $z = f(x, y)$, $x = x(u, v)$, $y = y(u, v)$ and $x(4, 5) = 2$, $y(4, 5) = 3$. Calculate the partial derivative in terms of $a, b, c, d, e, k, p, q, r, s, t, w$:

$$\begin{array}{llll} f_x(4, 5) = a & f_y(4, 5) = c & x_u(4, 5) = e & y_u(4, 5) = p \\ f_x(2, 3) = b & f_y(2, 3) = d & x_v(4, 5) = k & y_v(4, 5) = q \\ x_u(2, 3) = r & y_u(2, 3) = s & x_v(2, 3) = t & y_v(2, 3) = w \end{array}$$

35. $z_u(4, 5)$

36. $z_v(4, 5)$

For Problems 37–38, suppose that $x > 0, y > 0$ and that z can be expressed either as a function of Cartesian coordinates (x, y) or as a function of polar coordinates (r, θ) , so that $z = f(x, y) = g(r, \theta)$. [Recall that $x = r \cos \theta, y = r \sin \theta, r = \sqrt{x^2 + y^2}$, and, for $x > 0, y > 0, \theta = \arctan(y/x)$]

37. (a) Use the chain rule to find $\partial z/\partial r$ and $\partial z/\partial \theta$ in terms of $\partial z/\partial x$ and $\partial z/\partial y$.
 (b) Solve the equations you have just written down for $\partial z/\partial x$ and $\partial z/\partial y$ in terms of $\partial z/\partial r$ and $\partial z/\partial \theta$.
 (c) Show that the expressions you get in part (b) are the same as you would get by using the chain rule to find $\partial z/\partial x$ and $\partial z/\partial y$ in terms of $\partial z/\partial r$ and $\partial z/\partial \theta$.

38. Show that

$$\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 = \left(\frac{\partial z}{\partial r}\right)^2 + \frac{1}{r^2} \left(\frac{\partial z}{\partial \theta}\right)^2.$$

Problems 39–44 are continuations of the physical chemistry example on page 802.

39. Write $\left(\frac{\partial U}{\partial P}\right)_V$ as a partial derivative of one of the functions U_1, U_2 , or U_3 .
 40. Write $\left(\frac{\partial U}{\partial P}\right)_T$ as a partial derivative of one of the functions U_1, U_2, U_3 .
 41. For the gas in Example 6, find $\left(\frac{\partial U}{\partial T}\right)_V$ and $\left(\frac{\partial U}{\partial V}\right)_T$. [Hint: Use the same method as the example, but substitute for dP instead of dV .]
 42. Show that $\left(\frac{\partial T}{\partial V}\right)_P = 1 / \left(\frac{\partial V}{\partial T}\right)_P$.

43. Use Example 7 and Problem 42 to show that

$$\left(\frac{\partial U}{\partial V}\right)_P = \left(\frac{\partial U}{\partial V}\right)_T + \left(\frac{\partial U}{\partial T}\right)_V \frac{1}{\left(\frac{\partial T}{\partial V}\right)_P}.$$

44. In Example 6, we calculated values of $(\partial U/\partial T)_P$ and $(\partial U/\partial P)_T$ using the relationship $PV = 2T$ for a specific gas. In this problem, you will derive general relationships for these two partial derivatives.

- (a) Think of V as a function of P and T and write an expression for dV .
 (b) Substitute for dV into the following formula for dU (thinking of U as a function of P and V):

$$dU = \left(\frac{\partial U}{\partial P}\right)_V dP + \left(\frac{\partial U}{\partial V}\right)_P dV.$$

- (c) Thinking of U as a function of P and T , write an expression for dU .
 (d) By comparing coefficients of dP and dT in your answers to parts (b) and (c), show that

$$\begin{aligned} \left(\frac{\partial U}{\partial T}\right)_P &= \left(\frac{\partial U}{\partial V}\right)_P \cdot \left(\frac{\partial V}{\partial T}\right)_P \\ \left(\frac{\partial U}{\partial P}\right)_T &= \left(\frac{\partial U}{\partial P}\right)_V + \left(\frac{\partial U}{\partial V}\right)_P \cdot \left(\frac{\partial V}{\partial P}\right)_T. \end{aligned}$$

Problems 45–47 concern differentiating an integral in one variable, y , which also involves another variable x , either in the integrand, or in the limits, or both:

$$\int_0^5 (x^2 y + 4) dy \quad \text{or} \quad \int_0^x (y + 4) dy \quad \text{or} \quad \int_0^x (xy + 4) dy.$$

To differentiate the first integral with respect to x , it can be shown that in most cases we can differentiate with respect to x inside the integral:

$$\frac{d}{dx} \left(\int_0^5 (x^2 y + 4) dy \right) = \int_0^5 2xy dy.$$

Differentiating the second integral with respect to x uses the Fundamental Theorem of Calculus:

$$\frac{d}{dx} \int_0^x (y + 4) dy = x + 4.$$

Differentiating the third integral involves the chain rule, as shown in Problem 47. Assume that the function F is continuously differentiable and b is constant throughout.

45. Let $f(x) = \int_0^b F(x, y) dy$. Find $f'(x)$.
 46. Let $f(x) = \int_0^x F(b, y) dy$. Find $f'(x)$.
 47. Let $f(x) = \int_0^x F(x, y) dy$. Use Problem 45 and Problem 46 to find $f'(x)$ by the following steps:
 (a) Let $G(u, w) = \int_0^w F(u, y) dy$. Find $G_u(u, w)$ and $G_w(u, w)$.
 (b) Use part (a) and the chain rule applied to $G(x, x) = f(x)$ to show:

$$f'(x) = \int_0^x F_x(x, y) dy + F(x, x).$$

Strengthen Your Understanding

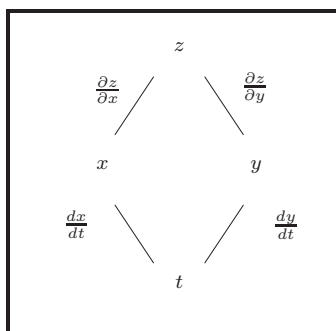
In Problems 48–50, explain what is wrong with the statement.

48. If $z = f(g(t), h(t))$, then $dz/dt = f(g'(t), h(t)) + f(g(t), h'(t))$.
49. If $C = C(R, T)$, $R = R(x, y)$, $T = T(x, y)$ and $R(0, 2) = 5$, $T(0, 2) = 1$, then $C_x(0, 2) = C_R(0, 2)R_x(0, 2) + C_T(0, 2)T_x(0, 2)$.
50. If $z = f(x, y)$ and $x = g(t)$, $y = h(t)$ with $g(0) = 2$ and $h(0) = 3$, then

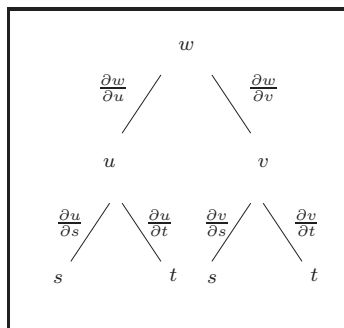
$$\left. \frac{dz}{dt} \right|_{t=0} = f_x(0, 0)g'(0) + f_y(0, 0)h'(0).$$

In Problems 51–55, give an example of:

51. Functions $x = g(t)$ and $y = h(t)$ such that $(dz/dt)|_{t=0} = 9$, given that $z = x^2y$.
52. A function $z = f(x, y)$ such that $dz/dt|_{t=0} = 10$, given that $x = e^{2t}$ and $y = \sin t$.
53. Functions z , x and y where you need to follow the diagram in order to answer questions about the derivative of z with respect to the other variables.



54. Functions w , u and v where you need to follow the diagram in order to answer questions about the derivative of w with respect to the other variables.



55. Function $z = f(x, y)$ where x and y are functions of one variable, t , for which $\frac{\partial z}{\partial t} = 2$.
56. Let $z = g(u, v)$ and $u = u(x, y, t)$, $v = v(x, y, t)$ and $x = x(t)$, $y = y(t)$. Then the expression for dz/dt has
- | | |
|-----------------|-----------------------|
| (a) Three terms | (b) Four terms |
| (c) Six terms | (d) Seven terms |
| (e) Nine terms | (f) None of the above |

14.7 SECOND-ORDER PARTIAL DERIVATIVES

Since the partial derivatives of a function are themselves functions, we can differentiate them, giving *second-order partial derivatives*. A function $z = f(x, y)$ has two first-order partial derivatives, f_x and f_y , and four second-order partial derivatives.

The Second-Order Partial Derivatives of $z = f(x, y)$

$$\begin{aligned} \frac{\partial^2 z}{\partial x^2} &= f_{xx} = (f_x)_x, & \frac{\partial^2 z}{\partial x \partial y} &= f_{yx} = (f_y)_x, \\ \frac{\partial^2 z}{\partial y \partial x} &= f_{xy} = (f_x)_y, & \frac{\partial^2 z}{\partial y^2} &= f_{yy} = (f_y)_y. \end{aligned}$$

It is usual to omit the parentheses, writing f_{xy} instead of $(f_x)_y$ and $\frac{\partial^2 z}{\partial y \partial x}$ instead of $\frac{\partial}{\partial y} \left(\frac{\partial z}{\partial x} \right)$.